

Speech-Adaptive Smart Glasses Teleprompter

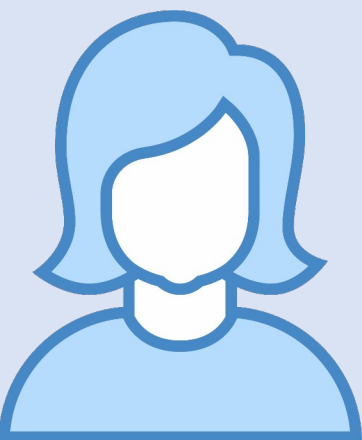
University at Albany · FACETS Lab / ATIC · NYS Chief Disability Office · OPWDD-supported research · Project team



The teleprompter follows the speaker.

Live speech → script alignment → adaptive pacing → readable smart-glasses text

Problem & Accessibility Need



Problem

I want to use a **teleprompter** during a **public speech** without making it obvious to the audience, so I need to read the script through a **glasses display**. But because of a **nerve condition**, I cannot reliably move my hand to control the display. I need the text to sit higher in my field of view and **scroll smoothly** on its own so I can focus on speaking instead of manually controlling the teleprompter.

Solution

Our goal is to make the glasses adapt to the user instead. The system could automatically control and update what is displayed based on the person's speech, pace, or other accessible inputs, reducing the need for manual interaction.



ATIC

Traditional vs. Our Approach

TRADITIONAL MODE

- Fixed-speed scrolling or manual control with the Neural Band
- The speaker follows the display.

OUR SPEECH-ADAPTIVE MODE

- Speaker's voice → adaptive scrolling
- Display follows speaker
- No Neural Band

Key Accessibility Features



Hands-free

Follows speech without continuous input.



Adaptive scroll

Adjusts to speaking pace.



Pause & stop

Let a speaker pause or stop progress.



Readable display

Improved clarity on the glasses.



Re-synchronization

Finds the script again after a deviation.



Personal pacing

Offers mode and speed preferences.

How It Works



PREPARED SCRIPT

Loaded in the phone app



LIVE SPEECH

Captured while presenting

1 Recognize

Speech to text

2 Align

Word + semantic match

3 Adapt

Pace + position bounds

4 Display

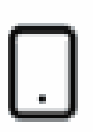
Smart-glasses text

Current System Architecture



Input & control

Speech recognition
Known script + alignment
Timer or voice mode



Phone application

Weights + rehearsal
Velocity + position bounds
Text / raster composition



Glasses display

DAT full-content delivery
600 × 600 additive view
Seven-line script window

Frame requests

Completion timing

Bluetooth sends

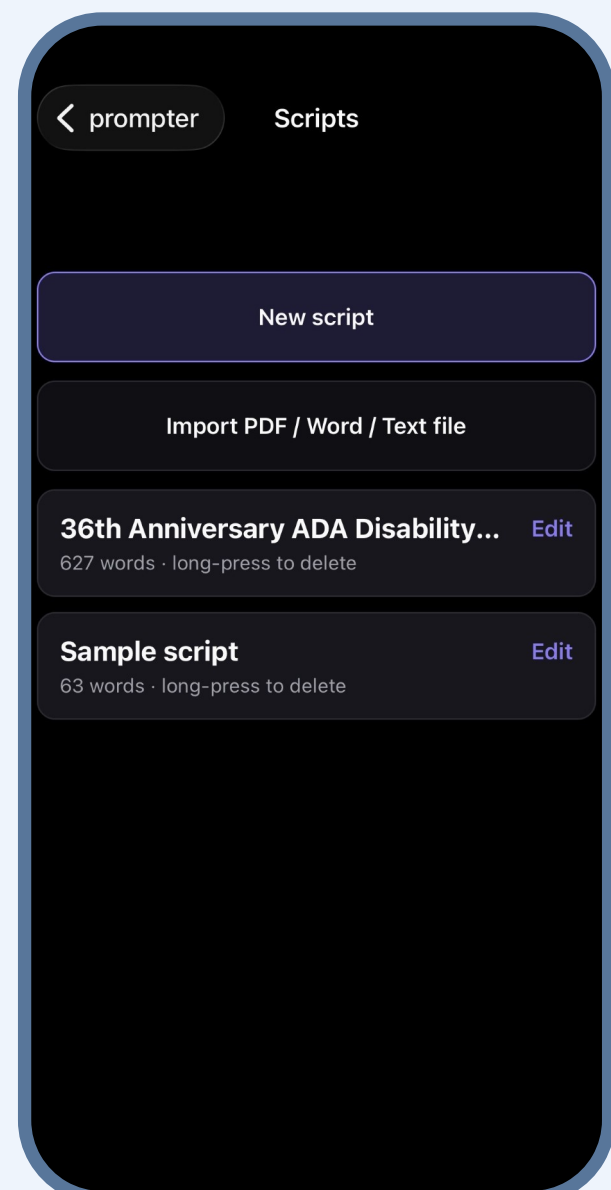


Send scheduler

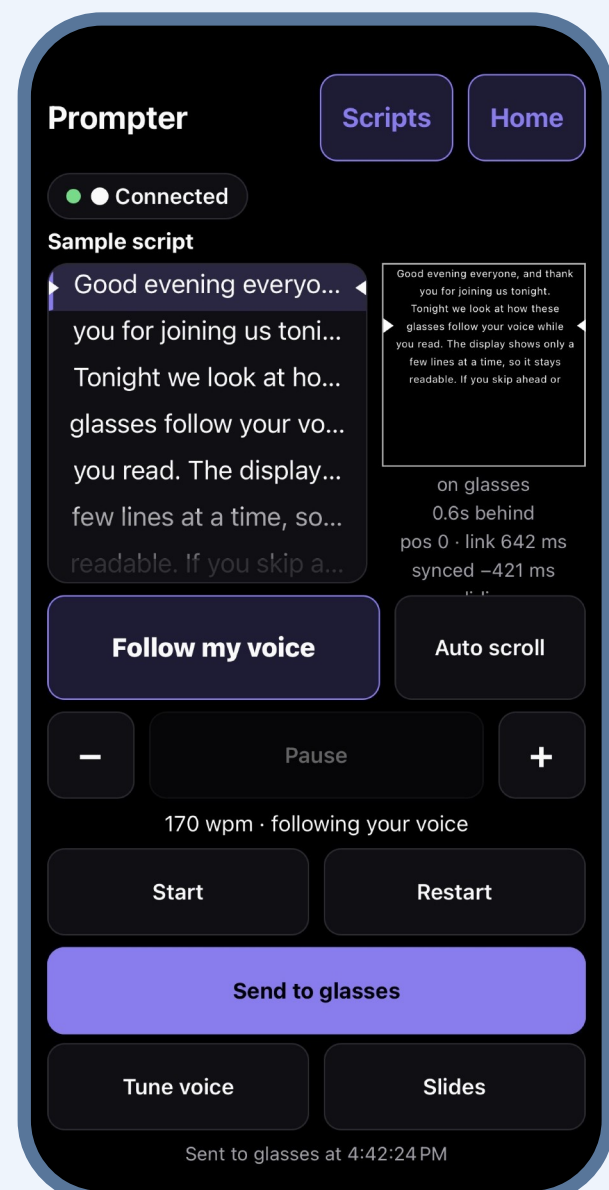
Probe + completion EWMA
Startup assumption: 2.5 Hz
Sustained rate unmeasured

App Interface & Glasses View

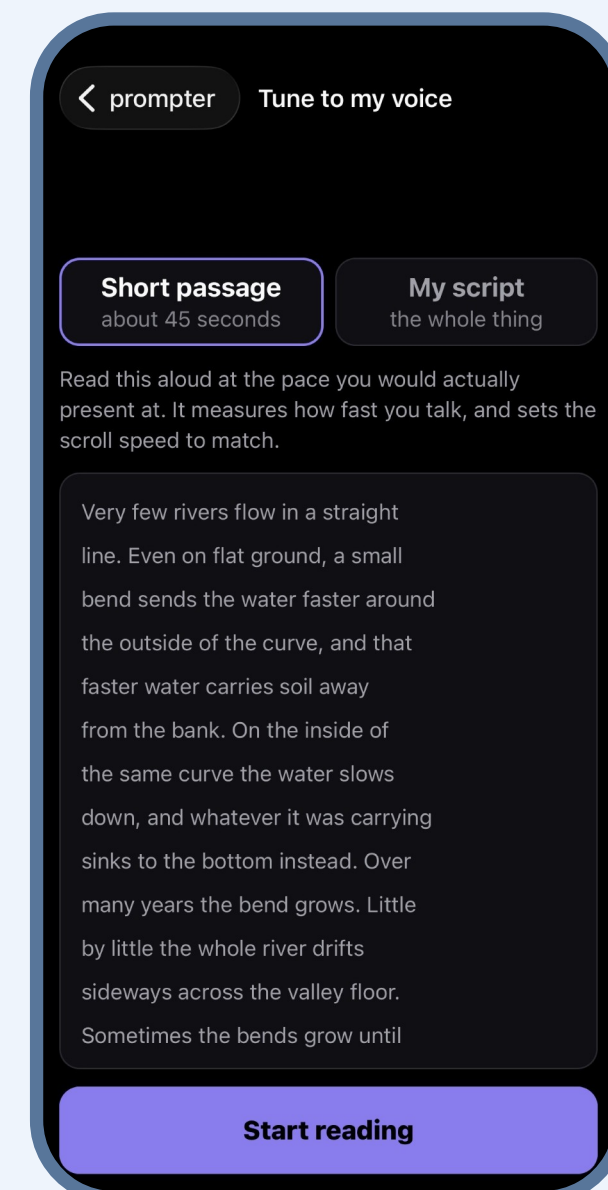
SCRIPT INPUT



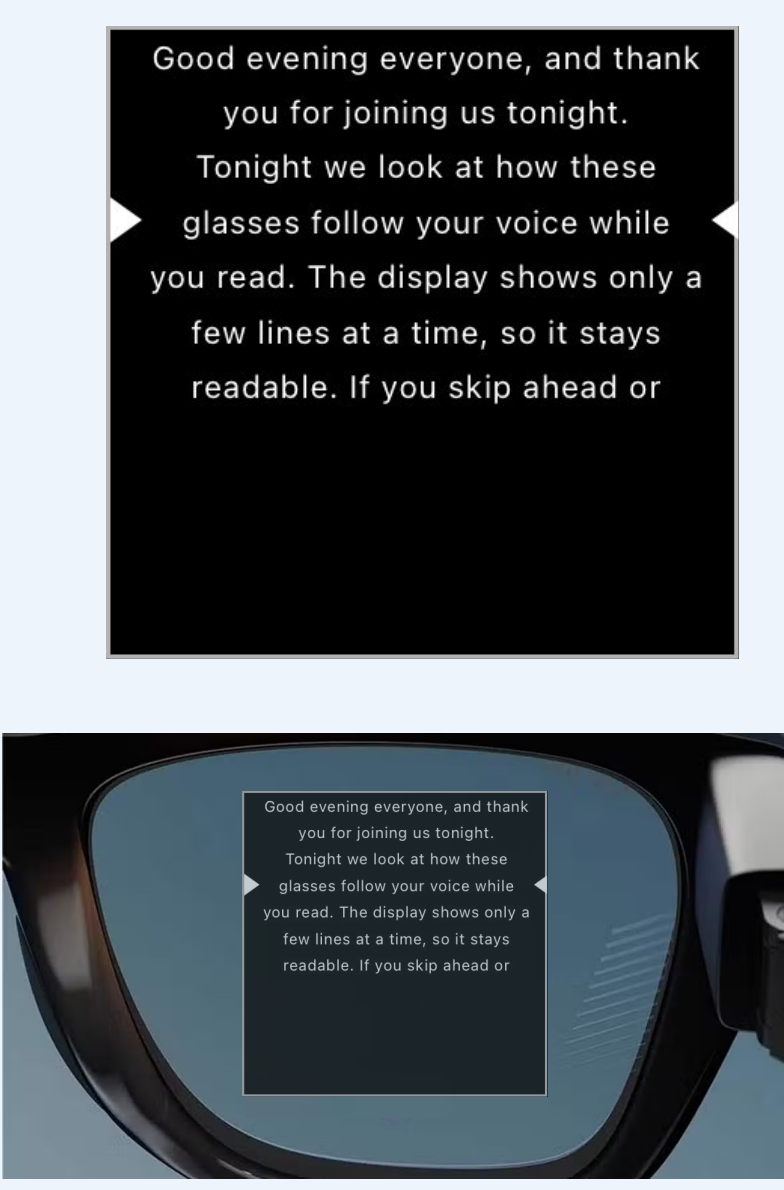
CONTROL PANEL



TUNE MY VOICE



GLASSES VIEW



Real-World Application & Collaboration

Collaboration: We tested the teleprompter app with the NYS Chief Disability Office from OPWDD while reducing the need for manual control.

Hardware challenge: The display sits low in the field of view. We are working with the Meta team to improve visibility.

Compatibility challenge: We tested other glasses, such as Meta Ray-Ban smart glasses and Even Realities G2, to make our app generalizable across different devices.



Meta

Even Realities

Technology & Current Progress

CURRENT HARDWARE

Meta Ray-Ban Display + mobile companion app

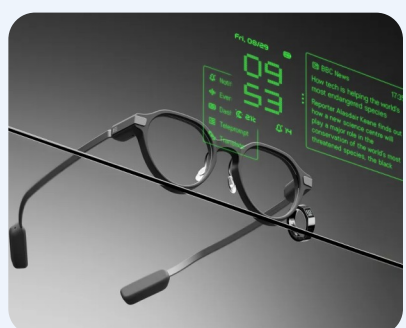
CORE METHODS

Real-time ASR · word-level alignment · semantic recovery · adaptive pacing · display rendering

REFINED THROUGH TESTING

Scroll behavior · clarity · stop function

Future Research: Multimodal Access



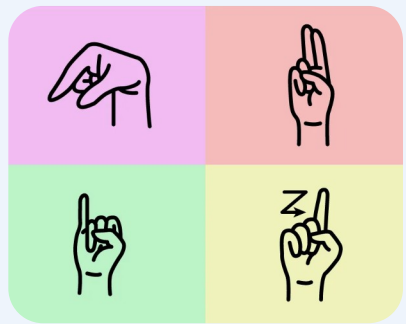
MULTI-DEVICE GLASSES

Testing Even Realities G2; developing an architecture for multiple displays.



ALTERNATIVE AUDIO & VOICE INTERFACES

Evaluating Shokz bone-conduction headsets for voice interaction and exploring their usability for people with conductive or mixed hearing loss and single-sided deafness.



SIGN LANGUAGE

Exploring future vision-based real-time, bidirectional translator between sign language and spoken language.

Video Demo & Vision



Toward wearable systems that adapt to the person across communication, sensory, and physical-access needs.